

Mohammad Reza Kianifar

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Skills

Programming Languages Python, Java, Kotlin, JavaScript, TypeScript, Bash, PHP, C++

Web Django, React, Spring Boot, Laravel, RPC, GraphQL, REST

Big Data & Databases PostgreSQL, MySQL, MongoDB, Kafka, Celery, Redis, Cassandra, Elastic Search

Site Reliability Engineering Docker, Kubernetes, GKE, AWS, Jenkins, GitHub Actions, Datadog, Circle CI

ML Engineering LLM Evaluation, Agentic Systems, Benchmarking, LLM-as-a-Judge, MCP

Software Engineering & Tools Microservices, Object-oriented patterns, Agile (Scrum), Git, Unit testing

Professional Experience

Senior Machine Learning Engineer Turing, Canada

Oct 2025 - Present

- Built a production platform on the **Harbor** framework for authoring, evaluating, and reviewing agentic coding tasks across benchmarks such as Terminal-Bench and SWE-Bench, with coding agents including Terminus-2, Claude Code, and Codex.
- Scaled the platform to 500+ monthly active users across 22 teams, running 300K+ evaluations and 130K+ verifications per month across 100K+ tasks.
- Designed a **Celery**-based concurrency scheduler with per-user limits and PostgreSQL advisory locks, running 1,000+ concurrent agentic trials without deadlocks.
- Designed a data-science benchmark harness that connects agents to **Jupyter** notebook environments via **MCP**, with rubric-based **LLM-as-a-judge** grading.
- Developed chat-based **agentic verification** with dynamic context selection that reviews and automatically fixes benchmark tasks, producing 4.5K+ corrected task revisions.
- Built **LLM-driven task generation** that extracts candidates from code repositories and produces benchmark-specific Harbor tasks for evaluation and training, generating 1.2K+ tasks from 400+ repositories.
- Created a visual **pipeline system** for designing, reusing, and automating task-review workflows, reaching 35K+ pipeline runs within its first month.

Software Engineer Wave HQ, Canada

Jan 2024 - Sep 2025

- Contributed to the risk management systems at **Wave**, a financial platform with 200K+ active small business users, using **Python**, **Django**, **PostgreSQL**, **RPC**, **GraphQL** and **React**.
- Led the architecture design and end-to-end implementation of Apple Pay integration.
- Developed an automated schema diagram generator for APIs and Kafka events used across teams.
- Established clean code and review principles to improve code quality across a 13-person engineering team.
- Reduced alert noise by 45% by setting up **Datadog** monitoring and observability conventions.

Google Summer of Code 2025 Contributor Zulip, Remote

May 2025 - Sep 2025

- Migrated **Zulip's** 1:1 and self-direct messages from the legacy **Persona1** recipient model to a unified **DirectMessageGroup** model, enabling per-conversation features such as pinning and hiding DMs.
- Merged **18 PRs** for a phased, backward-compatible migration across message sending, unread counts, notifications, scheduled messages, analytics, and Slack, Mattermost, and Rocket.Chat imports, using **Python**, **Django**, and **PostgreSQL**.
- Developed batched PostgreSQL backfill migrations for messages, drafts, scheduled messages, and archived messages to transition the recipient model with minimal downtime.

Graduate Research Assistant University of Calgary, Canada

Jan 2021 - Jan 2024

- Published a peer-reviewed journal article in **MDPI** on *Graph Generalization for Software Engineering*.
- Designed and implemented an algorithm to compare two directed labeled graphs to identify the similarities and abstract away differences with variable components with 90% test coverage.
- Detected semantic changes in library updates with 54% higher precision than existing tools.

Lead Software Engineer *Tapsell, Iran*

Jun 2020 - Dec 2020

- > **Tapsell** is the largest mobile ad network in Iran with 70M+ monthly active users and 100M+ daily requests.
- > Led the design of a video-ad integration with a **partner company** that handles 3M+ daily requests.
- > Managed a cross-functional team of 12 technical and communicated with business members for 6 months.
- > Developed a comprehensive documentation platform, saving 25+ hours of training time per intern.
- > Supervised a 2-week internship program for college students with hands-on exercises.

Software Engineer *Tapsell, Iran*

Jan 2019 - Jun 2020

- > Created a data collector module for an integrated service, handling 5M+ daily requests. **Kafka**, **Cassandra**, and **Elasticsearch** were the main components.
- > Collaborated on 12 projects in a microservice architecture involving designing data pipelines and managing high volumes of requests, using **Java**, **Kotlin**, and **Spring Boot** deployed with **Jenkins** and **Kubernetes**.
- > Facilitated the streaming processing service, which replaced the previous Spark-based version.
- > Designed new services based on RESTful API for the dashboard panel with 55% improvements in performance and resources, by optimizing **PostgreSQL** and **MongoDB** databases.
- > Refactored and optimized the code base, improving code readability and performance using clean code principles by removing 4000+ lines of legacy code.

Software Engineer *Hinza Co., Iran*

Jan 2016 - Dec 2018

- > **Hinza** provides digital solutions, especially website services, for 50+ medium-sized companies.
- > Automated time-tracking for employees by developing an online application with a test coverage of 95%. Coded the hybrid application using **React Native** and backend using **Python**, **Django**, and **MySQL**.
- > Played an integral role in the development of an E-Commerce platform, leading to the sale of profitable projects, notably **Kitline**, using **JavaScript**, **PHP**, and **Laravel**.
- > Collaborated on the development of the company's content management system used on 20+ websites.

Education

Master of Science in Computer Science *University of Calgary, Canada (GPA 3.93/4)*

Jan 2021 - Jan 2024

Bachelor of Science in Software Engineering *University of Tehran, Iran (GPA 3.79/4)*

Sep 2015 - Feb 2020

Volunteer Activities

President, Computer Science Graduate Society *University of Calgary*

Jun 2021 - Apr 2023

- > Raised \$10K+ in funding, held 10+ events, and received the \$3,000 Rizvi Family Scholarship.

Chief Teaching Assistant, Internet Engineering *University of Tehran*

Jan 2019 - May 2019

- > Led the design of 10 assignments for 80 students while supervising a team of 9 TAs.

Technical Mentor, Summer of Code *University of Tehran*

Jul 2018 - Sep 2018

- > Coordinated a team of 5 students to build a social network for university events using **NodeJS** and **React**.

Awards

Turing Google Hero *Monthly recognition for outstanding contributions at Turing*

Dec 2025

CCPC, 4th Place *Data structures and algorithms contest, two consecutive years*

Apr 2022, Apr 2023

Iranian University Entrance Exam *Ranked 364th among 180K+ (top 0.2%)*

Jul 2015